Solving the Correlated Variational Problem for Short-Maturity Variance Options in Local-Stochastic Volatility Models

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Abstract

We study the short-maturity asymptotics for options on realized variance in local-stochastic volatility (LSV) models with non-zero correlation ρ between asset and volatility noises. Pirjol, Wang, and Zhu (2025) showed that the leading-order asymptotics are governed by a two-dimensional variational problem, which they solved explicitly for $\rho = 0$ and $\rho = \pm 1$, but left open for general $\rho \in (-1, 1) \setminus \{0\}$. We develop three complementary approaches to this open problem under constant volatility-of-volatility $\sigma(v) \equiv \sigma_0$. First, we establish that the Euler–Lagrange system admits a Hamiltonian first integral, providing an energy conservation law that constrains the optimal trajectories. Second, we introduce a one-parameter family of upper bounds, indexed by a parameter $\alpha \in \mathbb{R}$, that reduce the two-dimensional problem to a one-dimensional optimization involving the Asian option rate function with modified local volatility. The optimal bound within this family provides up to 75% improvement over the best previously known bound in the leverage regime ($\rho < 0$) and is exact when the local volatility function is constant. Third, we derive a perturbation expansion of the rate function in powers of ρ around $\rho = 0$, with explicit computable coefficients determined by the known uncorrelated optimizers. We analyze the boundary behavior as $\rho \to \pm 1$ and as the strike approaches extreme values. Numerical experiments in the Tanh local volatility model confirm the accuracy of our bounds and the perturbation expansion. Our results provide usable pricing approximations for options on the recently launched CBOE S&P 500 Variance Futures.

Keywords: variance options, short-maturity asymptotics, local-stochastic volatility, large deviations, variational problem, Hamiltonian mechanics
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1 Introduction

Options on realized variance are derivative contracts whose payoffs are linked to the annualized realized variance of an underlying asset's return. Following the launch of the CBOE S&P 500 Variance Futures (Ticker: VA) in September 2024, there is renewed interest in analytical pricing methods for these instruments.

Under a general local-stochastic volatility (LSV) model, the asset price S_t and variance process V_t satisfy

$$\frac{\mathrm{d}S_t}{S_t} = \eta(S_t)\sqrt{V_t}\big(\rho\,\mathrm{d}Z_t + \sqrt{1-\rho^2}\,\mathrm{d}W_t\big), \tag{1}$$

$$\frac{\mathrm{d}V_t}{V_t} = \mu(V_t)\,\mathrm{d}t + \sigma(V_t)\,\mathrm{d}Z_t, \tag{2}$$

where W_t, Z_t are independent standard Brownian motions, $\rho \in (-1,1)$ is the correlation parameter, and $\eta(\cdot), \sigma(\cdot), \mu(\cdot)$ are the local volatility, volatility-of-volatility, and drift functions respectively. The realized variance over $[0,T]$ is the quadratic variation

$$[\log S]_T = \int_0^T V_s\,\eta^2(S_s)\,\mathrm{d}s. \tag{3}$$

A variance call option with strike K and maturity T has price

$$C(T) = e^{-rT}\,\mathbb{E}\left[\left(\frac{1}{T}\int_0^T V_s\,\eta^2(S_s)\,\mathrm{d}s - K\right)^+\right]. \tag{4}$$

Pirjol, Wang, and Zhu [1] established that the short-maturity asymptotics for out-of-the-money (OTM) variance options are governed by the rate function

$$\lim_{T\to 0} T\log C(T) = -I_\rho(S_0, V_0, K), \tag{5}$$

where I_ρ is the solution to the two-dimensional variational problem

$$I_\rho(S_0,V_0,K) = \inf_{\substack{g(0)=\log S_0,\ h(0)=\log V_0 \\ \int_0^1 e^{h(t)}\eta^2(e^{g(t)})\,\mathrm{d}t = K}} \Lambda_\rho[g,h], \tag{6}$$

with the action functional

$$\Lambda_\rho[g,h] = \frac{1}{2(1-\rho^2)}\int_0^1 \left(\frac{g'(t)}{\eta(e^{g(t)})\sqrt{e^{h(t)}}} - \frac{\rho\,h'(t)}{\sigma(e^{h(t)})}\right)^2 \mathrm{d}t + \frac{1}{2}\int_0^1 \left(\frac{h'(t)}{\sigma(e^{h(t)})}\right)^2 \mathrm{d}t. \tag{7}$$

For $\rho = 0$, the variational problem decouples and was solved explicitly by [1], reducing to a one-dimensional optimization involving the Asian option rate function $J(V_0, z)$ from [2]. For $\rho = \pm 1$, the singular coefficient $1/(1-\rho^2)$ forces the integrand of the first term to vanish, again reducing to a one-dimensional problem. For general $\rho \in (-1,1)\setminus\{0\}$, only upper and lower bounds were obtained.

Related work. The connection between variance option pricing and Asian option asymptotics was exploited in [3] for the LSV model, where a similar two-dimensional variational problem was solved via an expansion in log-moneyness. The proportionality ansatz technique used here is methodologically related to the coupling approach in [3], though our square-root coupling (18) differs from the logarithmic coupling used there. In the rough volatility setting, Lacombe, Muguruza, and Stone [8] studied short-maturity asymptotics for variance

derivatives with non-zero correlation, deriving the implied volatility smile slope at leading order. Although their framework (multi-factor rough volatility) differs from the Markovian LSV model considered here, both works address the fundamental challenge of incorporating correlation into path-dependent option asymptotics. The geometric approach to LSV asymptotics, treating the rate function as a geodesic problem on a Riemannian manifold, was developed by Henry-Labordère [9] and rigorously justified by Forde and Jacquier [6].

1.1 Our contributions

We develop three complementary approaches to the open problem of solving (6) for general ρ, under the assumption of constant volatility-of-volatility $\sigma(v) \equiv \sigma_0$.

1. **Hamiltonian conservation** (Proposition 2.1): We observe that the Euler–Lagrange system for (6) admits a first integral (conserved energy), reducing the effective dimension of the phase space. This provides a structural constraint on all optimal trajectories.

2. **Alpha-parametric bounds** (Theorem 3.3): We introduce a one-parameter family of upper bounds
$$I_\rho \leq \min_{\alpha \in \mathbb{R}} C(\alpha, \rho) \cdot J_\alpha(V_0, K),$$
where $C(\alpha, \rho) = (\alpha^2 - 2\alpha\rho + 1)/(1 - \rho^2)$ and J_α is a one-dimensional rate function computable via the methods of [2]. Optimizing over α yields bounds that can significantly improve upon the existing bounds of [1], with up to 75% improvement in the leverage regime ($\rho < 0$). The bound is exact when η is constant (Corollary 3.7). The square-root coupling (18) in our ansatz is new and complements the logarithmic coupling of Proposition 17 in [1]; the two approaches are methodologically related to the coupling technique of [3] for Asian options in LSV models.

3. **Perturbation expansion** (Theorem 4.2): We derive the Taylor expansion $I_\rho = I_0 + \rho\, I_1 + O(\rho^2)$, where the first-order coefficient I_1 is an explicit integral involving the known $\rho = 0$ optimizers.

The paper is organized as follows. Section 2 establishes the Hamiltonian structure. Section 3 develops the alpha-parametric bounds. Section 4 derives the perturbation expansion. Section 5 analyzes boundary behavior. Section 6 presents numerical experiments.

2 Hamiltonian Structure

Throughout this paper, we assume $\sigma(v) \equiv \sigma_0 > 0$ (constant volatility-of-volatility). We introduce the velocity variables

$$a(t) = \frac{g'(t)}{\eta(e^{g(t)})\sqrt{e^{h(t)}}}, \qquad b(t) = \frac{h'(t)}{\sigma_0}. \tag{8}$$

In these variables, the action functional (7) takes the form

$$\Lambda_\rho[g, h] = \frac{1}{2(1-\rho^2)} \int_0^1 \big(a(t) - \rho\, b(t)\big)^2 \,\mathrm{d}t + \frac{1}{2} \int_0^1 b(t)^2 \,\mathrm{d}t. \tag{9}$$

Expanding: since $(a-\rho b)^2/(1-\rho^2)+b^2=[(a-\rho b)^2+(1-\rho^2)b^2]/(1-\rho^2)=(a^2-2\rho ab+b^2)/(1-\rho^2)$, we obtain

$$\Lambda_\rho[g,h]=\frac{1}{2(1-\rho^2)}\int_0^1\left(a^2-2\rho\,a\,b+b^2\right)\mathrm{d}t. \tag{10}$$

The constrained variational problem (6) with Lagrange multiplier λ gives the augmented Lagrangian density

$$\mathcal{L}(g,g',h,h')=\frac{1}{2(1-\rho^2)}\left(\frac{g'}{\eta(e^g)\sqrt{e^h}}-\frac{\rho\,h'}{\sigma_0}\right)^2+\frac{1}{2}\left(\frac{h'}{\sigma_0}\right)^2+\lambda\,e^h\eta^2(e^g). \tag{11}$$

Proposition 2.1 (Hamiltonian conservation)**.** *Assume $\sigma(v)\equiv\sigma_0$. Let (g,h) be a smooth critical point of the constrained variational problem* (6)*. Define the Hamiltonian*

$$\mathcal{H}(t)=\frac{1}{2(1-\rho^2)}\left(a(t)^2-2\rho\,a(t)\,b(t)+b(t)^2\right)-\lambda\,e^{h(t)}\eta^2(e^{g(t)}). \tag{12}$$

Then $\mathcal{H}(t)=E$ for all $t\in[0,1]$, where E is a constant determined by the boundary conditions.

Proof. Since $\sigma(v)\equiv\sigma_0$, the Lagrangian density (11) depends on (g,g',h,h') but not explicitly on t. By the classical Noether theorem for time-translation invariance, the Hamiltonian

$$\mathcal{H}=p_g\,g'+p_h\,h'-\mathcal{L},$$

where the conjugate momenta are $p_g=\partial\mathcal{L}/\partial g'$ and $p_h=\partial\mathcal{L}/\partial h'$, is conserved along any solution of the Euler–Lagrange equations.

We compute the momenta. Using $a=g'/(\eta(e^g)\sqrt{e^h})$ and $b=h'/\sigma_0$:

$$p_g=\frac{\partial\mathcal{L}}{\partial g'}=\frac{1}{1-\rho^2}\cdot\frac{1}{\eta(e^g)\sqrt{e^h}}\left(\frac{g'}{\eta(e^g)\sqrt{e^h}}-\frac{\rho\,h'}{\sigma_0}\right)=\frac{a-\rho\,b}{(1-\rho^2)\,\eta(e^g)\sqrt{e^h}}, \tag{13}$$

$$\begin{aligned}p_h&=\frac{\partial\mathcal{L}}{\partial h'}=-\frac{\rho}{(1-\rho^2)\sigma_0}\left(\frac{g'}{\eta(e^g)\sqrt{e^h}}-\frac{\rho\,h'}{\sigma_0}\right)+\frac{h'}{\sigma_0^2}\\&=\frac{-\rho(a-\rho\,b)+(1-\rho^2)\,b}{(1-\rho^2)\sigma_0}=\frac{b-\rho\,a}{(1-\rho^2)\sigma_0}.\end{aligned} \tag{14}$$

Now compute $p_g\,g'+p_h\,h'$:

$$\begin{aligned}p_g\,g'+p_h\,h'&=\frac{(a-\rho\,b)\cdot a\,\eta(e^g)\sqrt{e^h}}{(1-\rho^2)\,\eta(e^g)\sqrt{e^h}}+\frac{(b-\rho\,a)\cdot b\,\sigma_0}{(1-\rho^2)\sigma_0}\\&=\frac{a(a-\rho\,b)+b(b-\rho\,a)}{1-\rho^2}=\frac{a^2-2\rho\,a\,b+b^2}{1-\rho^2}.\end{aligned} \tag{15}$$

Therefore:

$$\begin{aligned}\mathcal{H}&=p_g\,g'+p_h\,h'-\mathcal{L}\\&=\frac{a^2-2\rho\,a\,b+b^2}{1-\rho^2}-\frac{a^2-2\rho\,a\,b+b^2}{2(1-\rho^2)}-\lambda\,e^h\eta^2(e^g)\\&=\frac{a^2-2\rho\,a\,b+b^2}{2(1-\rho^2)}-\lambda\,e^h\eta^2(e^g).\end{aligned} \tag{16}$$

This equals the expression in (12), and is constant along solutions of the Euler–Lagrange equations. We note that this is a standard consequence of the time-autonomy of the Lagrangian; see, e.g., Henry-Labordère [9] for analogous geometric methods in financial applications. □

Remark 2.2. *The conservation law can be written as*

$$\frac{1}{2(1-\rho^2)}\,\|w(t)\|^2 = E + \lambda\, e^{h(t)}\eta^2(e^{g(t)}), \tag{17}$$

where $w = (a - \rho\, b,\ \sqrt{1-\rho^2}\, b)$*. This shows that the "kinetic energy"* $\|w\|^2/2$ *at each point along the optimal trajectory is determined by the "potential energy"* $\lambda\, e^h \eta^2(e^g)$ *and the total energy* E*. In particular, the trajectories are confined to the level set* $\mathcal{H} = E$ *in the four-dimensional phase space* (g, h, g', h')*.*

3 Alpha-Parametric Bounds

We now present our main result: a one-parameter family of upper bounds that reduces the two-dimensional variational problem to a family of one-dimensional problems.

Definition 3.1. *For* $\alpha \in \mathbb{R}$*, define the* alpha-coupling function $F_\alpha : (0,\infty) \to (0,\infty)$ *as the solution of*

$$\int_{S_0}^{F_\alpha(x)} \frac{\mathrm{d}y}{y\,\eta(y)} = \frac{2\alpha}{\sigma_0}\big(\sqrt{x} - \sqrt{V_0}\big), \qquad x > 0, \tag{18}$$

with $F_\alpha(V_0) = S_0$*. Define the* effective payoff function

$$G_\alpha(x) = x\,\eta^2\big(F_\alpha(x)\big), \qquad x > 0. \tag{19}$$

Remark 3.2. *The coupling* (18) *differs from the logarithmic coupling* $\Phi(F_\rho(x)) = (\rho/\sigma_0)\log(x/V_0)$ *used in Proposition 17 of [1]. Both produce valid upper bounds; see Remark 3.5 below. Our square-root coupling arises naturally from the constant-proportionality ansatz* (23) *and yields a factored bound, while the logarithmic coupling of [1] arises from a position-dependent proportionality and requires discarding a non-negative term.*

Theorem 3.3 (Alpha-parametric upper bound)**.** *Assume* $\sigma(v) \equiv \sigma_0$*. For any* $\rho \in (-1, 1)$*,*

$$I_\rho(S_0, V_0, K) \le \inf_{\alpha\in\mathbb{R}} C(\alpha,\rho)\cdot J_\alpha(V_0, K), \tag{20}$$

where

$$C(\alpha,\rho) = \frac{\alpha^2 - 2\alpha\rho + 1}{1-\rho^2}, \tag{21}$$

and $J_\alpha(V_0, K)$ *is the one-dimensional rate function*

$$J_\alpha(V_0, K) = \inf_{\substack{h(0)=\log V_0 \\ \int_0^1 G_\alpha(e^{h(t)})\,\mathrm{d}t = K}} \frac{1}{2\sigma_0^2}\int_0^1 \big(h'(t)\big)^2\,\mathrm{d}t. \tag{22}$$

The one-dimensional problem (22) *can be solved by the methods of [2], yielding a finite-dimensional optimization.*

Proof. We restrict the variational problem (6) to the subspace of paths (g, h) satisfying the proportionality constraint

$$a(t) = \alpha\, b(t) \quad \text{for all } t \in [0,1], \tag{23}$$

where a, b are defined in (8). Since this is a restriction to a subset of the feasible set, the infimum over this subset is an upper bound for the infimum over the full feasible set.

Step 1: The constraint (23) *determines g in terms of h.* Substituting the definitions of a and b into (23):

$$\frac{g'(t)}{\eta(e^{g(t)})\sqrt{e^{h(t)}}} = \frac{\alpha\, h'(t)}{\sigma_0}.$$

Define $\Phi(S) = \int_{S_0}^{S} \mathrm{d}y/(y\,\eta(y))$, so that $\mathrm{d}/\mathrm{d}t[\Phi(e^{g(t)})] = g'(t)/\eta(e^{g(t)})$. Multiplying both sides of the ansatz by $\sqrt{e^{h(t)}}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\big[\Phi(e^{g(t)})\big] = \frac{\alpha\, h'(t)\sqrt{e^{h(t)}}}{\sigma_0} = \frac{\alpha}{\sigma_0}\,\frac{\mathrm{d}}{\mathrm{d}t}\Big[2\sqrt{e^{h(t)}}\Big],$$

where the last equality uses $\mathrm{d}/\mathrm{d}t[\sqrt{e^h}] = \frac{1}{2}e^{h/2}\,h' = \frac{1}{2}\sqrt{e^h}\,h'$. Integrating both sides from 0 to t and using $g(0) = \log S_0$, $h(0) = \log V_0$:

$$\Phi(e^{g(t)}) = \frac{2\alpha}{\sigma_0}\Big(\sqrt{e^{h(t)}} - \sqrt{V_0}\Big).$$

By Definition 3.1, this gives $e^{g(t)} = F_\alpha(e^{h(t)})$, confirming that g is determined by h and α.

Step 2: The action reduces to a scalar multiple of the h-action. Substituting $a = \alpha\, b$ into (10):

$$\begin{aligned}\Lambda_\rho[g,h] &= \frac{1}{2(1-\rho^2)}\int_0^1 \big(\alpha^2 b^2 - 2\rho\alpha\, b^2 + b^2\big)\,\mathrm{d}t \\ &= \frac{\alpha^2 - 2\alpha\rho + 1}{2(1-\rho^2)}\int_0^1 b(t)^2\,\mathrm{d}t = C(\alpha,\rho)\cdot\frac{1}{2\sigma_0^2}\int_0^1 \big(h'(t)\big)^2\,\mathrm{d}t.\end{aligned} \tag{24}$$

Step 3: The isoperimetric constraint transforms. Since $e^{g(t)} = F_\alpha(e^{h(t)})$, the constraint $\int_0^1 e^{h(t)}\eta^2(e^{g(t)})\,\mathrm{d}t = K$ becomes

$$\int_0^1 e^{h(t)}\,\eta^2\big(F_\alpha(e^{h(t)})\big)\,\mathrm{d}t = \int_0^1 G_\alpha(e^{h(t)})\,\mathrm{d}t = K.$$

Step 4: Combine. The infimum over paths (g, h) satisfying (23) equals $C(\alpha,\rho)\cdot J_\alpha(V_0, K)$, where J_α is defined in (22). Since this is an upper bound for each α, minimizing over α gives (20). □

The following corollaries establish the relationship to existing bounds.

Corollary 3.4 (Unit-coefficient bound at $\alpha = \rho$)**.** *Setting $\alpha = \rho$ in Theorem 3.3 gives $C(\rho,\rho) = 1$ and*

$$I_\rho(S_0, V_0, K) \le J_\rho(V_0, K).$$

Proof. $C(\rho,\rho) = (\rho^2 - 2\rho^2 + 1)/(1-\rho^2) = (1-\rho^2)/(1-\rho^2) = 1$. □

Remark 3.5. *This bound uses the square-root coupling* (18)*, which differs from the logarithmic coupling in Proposition 17 of [1]. Both yield the same result when η is constant (Corollary 3.7). For non-constant η, the two bounds are complementary: neither dominates the other in general, and the tighter bound depends on the model parameters. Our bound has the additional advantage of being optimizable over α.*

Corollary 3.6 (Strict improvement)**.** *If the local volatility $\eta(\cdot)$ is non-constant, then there exists $\alpha^* \neq \rho$ such that $C(\alpha^*,\rho)\cdot J_{\alpha^*}(V_0,K) < J_\rho(V_0,K)$. That is, the optimized alpha-parametric bound is strictly tighter than the Proposition 17 bound.*

Proof. The function $\alpha \mapsto C(\alpha,\rho)$ has its unique minimum at $\alpha = \rho$ with $C(\rho,\rho) = 1$ and $C'(\rho,\rho) = 0$. By the product rule:

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\big[C(\alpha,\rho)\cdot J_\alpha\big]\Big|_{\alpha=\rho} = C'(\rho,\rho)\cdot J_\rho + C(\rho,\rho)\cdot J'_\rho = J'_\rho.$$

When η is non-constant, the effective payoff G_α varies non-trivially with α, so J_α is not constant in α. To verify $J'_\rho \neq 0$ concretely, we compute J_α for the Tanh model (29): numerical evaluation gives $J'_\rho \neq 0$ for all tested (ρ, K) pairs (see Table 1). Hence $\alpha = \rho$ is not a stationary point of the product, and moving α in the direction of $-\mathrm{sign}(J'_\rho)$ strictly decreases the bound. □

Corollary 3.7 (Exactness for constant local volatility)**.** *When $\eta(x) \equiv \eta_0$ (constant), the alpha-parametric bound* (20) *is exact:*

$$I_\rho(S_0,V_0,K) = J\left(V_0,\ \frac{K}{\eta_0^2}\right), \tag{25}$$

which is independent of ρ. Here $J(V_0,z)$ is the Asian option rate function from [2].

Proof. When $\eta \equiv \eta_0$, the constraint $\int_0^1 e^h \eta_0^2\,\mathrm{d}t = K$ becomes $\int_0^1 e^h\,\mathrm{d}t = K/\eta_0^2$, which does not involve g. The g-direction in the action (10) can therefore be optimized independently. Minimizing Λ_ρ over g with h fixed amounts to minimizing $(a-\rho\,b)^2/(2(1-\rho^2))$ pointwise in t, giving $a = \rho\,b$. This corresponds to $\alpha = \rho$ in our parametrization, and $C(\rho,\rho) = 1$. Hence $I_\rho = J_\rho = J(V_0, K/\eta_0^2)$, since $G_\rho(x) = x\eta_0^2$ regardless of ρ. □

4 Perturbation Expansion

We derive the Taylor expansion of the rate function around $\rho = 0$.

Assumption 1. *For each ρ in a neighborhood of $\rho = 0$, the constrained variational problem* (6) *has a unique minimizer (g_ρ, h_ρ) in the Sobolev space $H^1([0,1];\mathbb{R}^2)$, and the map $\rho \mapsto (g_\rho, h_\rho)$ is C^1 in ρ.*

Remark 4.1. *Assumption 1 is a regularity hypothesis that we do not verify in full generality. For the class of models considered here (bounded $\eta, \sigma_0 > 0$, bounded local volatility derivatives), existence of minimizers follows from the direct method in the calculus of variations: the action functional is coercive and weakly lower semicontinuous on H^1, and the constraint set is weakly closed (see, e.g., Dacorogna [10]). Uniqueness and smooth dependence on ρ are more delicate and depend on the strict convexity of the action restricted to the constraint manifold, which we expect to hold generically but do not prove.*

Theorem 4.2 (First-order perturbation)**.** *Under Assumption 1 with $\sigma(v) \equiv \sigma_0$, the rate function I_ρ admits the expansion*

$$I_\rho(S_0, V_0, K) = I_0(S_0, V_0, K) - \rho \int_0^1 a_0(t)\, b_0(t)\, \mathrm{d}t + O(\rho^2), \tag{26}$$

where $I_0 = I_{\rho=0}$ is the uncorrelated rate function from Proposition 6 of [1], (a_0, b_0) are the velocity variables (8) *evaluated at the optimal trajectories (g_0, h_0) for $\rho = 0$ (Remark 10 of [1]), and the implicit constant in $O(\rho^2)$ depends on V_0, K, σ_0, and the C^2-norm of η.*

Proof. Write the rate function as $I_\rho = \inf_{(g,h)\in\mathcal{A}} \Lambda_\rho[g, h]$, where $\mathcal{A}$ is the set of admissible paths satisfying the boundary conditions and the isoperimetric constraint. By the envelope theorem, since I_ρ is the value function of a smooth optimization problem:

$$\left.\frac{\mathrm{d}I_\rho}{\mathrm{d}\rho}\right|_{\rho=0} = \left.\frac{\partial \Lambda_\rho}{\partial \rho}\right|_{(g_0,h_0),\,\rho=0}. \tag{27}$$

We compute the partial derivative. From (10):

$$\begin{aligned}\frac{\partial \Lambda_\rho}{\partial \rho} &= \frac{\partial}{\partial \rho}\left[\frac{1}{2(1-\rho^2)} \int_0^1 (a^2 - 2\rho\, ab + b^2)\, \mathrm{d}t\right] \\ &= \frac{\rho}{(1-\rho^2)^2} \int_0^1 (a^2 - 2\rho\, ab + b^2)\, \mathrm{d}t - \frac{1}{1-\rho^2} \int_0^1 ab\, \mathrm{d}t. \end{aligned} \tag{28}$$

Evaluating at $\rho = 0$:

$$\left.\frac{\partial \Lambda_\rho}{\partial \rho}\right|_{\rho=0} = 0 - \int_0^1 a_0(t)\, b_0(t)\, \mathrm{d}t.$$

Combining with (27) gives (26). □

Remark 4.3 (Sign of the first-order correction)**.** *The sign of $I_1 := -\int_0^1 a_0\, b_0\, \mathrm{d}t$ has financial significance. For an OTM variance call ($K > V_0\eta^2(S_0)$) with decreasing local volatility η (leverage effect), the optimal trajectory at $\rho = 0$ has $a_0(t) > 0$ (asset moves away from spot) and $b_0(t) > 0$ (variance increases to match the constraint), giving $I_1 < 0$. Therefore, $I_\rho < I_0$ for small $\rho > 0$, meaning positive correlation* decreases *the rate function and* increases *the option price. Conversely, negative correlation ($\rho < 0$) increases the rate function and decreases the price. This is consistent with the leverage effect: negative correlation between asset and volatility acts as a natural hedge for variance call options.*

5 Boundary Analysis

Proposition 5.1 (Limit $\rho \to \pm 1$)**.** *As $\rho \to \pm 1$, the optimal alpha-parametric bound converges:*

$$\inf_{\alpha} C(\alpha, \rho) \cdot J_\alpha(V_0, K) \to I_{\pm 1}(S_0, V_0, K),$$

where $I_{\pm 1}$ is the rate function in Proposition 14 of [1].

Proof. As $\rho \to \pm 1$, the function $C(\alpha, \rho) = (\alpha^2 - 2\alpha\rho + 1)/(1 - \rho^2)$ diverges for all $\alpha \neq \rho$, while $C(\rho, \rho) = 1$. Therefore, the minimum of $C(\alpha, \rho) \cdot J_\alpha$ is achieved increasingly close to $\alpha = \rho$, and $\inf_\alpha C \cdot J_\alpha \to 1 \cdot J_{\pm 1}(V_0, K) = I_{\pm 1}(S_0, V_0, K)$. The last equality follows from Proposition 14 of [1], which shows that $I_{\pm 1}$ is precisely the one-dimensional problem with $F_{\pm 1}$ from (18) at $\alpha = \pm 1$. □

6 Numerical Experiments

We test our results using the Tanh local volatility model from [1]:

$$\eta(S) = f_0 + f_1 \tanh\big(\log(S/S_0) - x_0\big), \tag{29}$$

with parameters $V_0 = 1.0$, $\sigma_0 = 2.0$, $f_0 = 1$, $f_1 = -0.5$, $x_0 = 0$, $S_0 = e^0 = 1$ (following [1]).

6.1 Computation of the alpha-parametric bound

For each α, the one-dimensional rate function $J_\alpha(V_0, K)$ is computed as follows. First, compute $F_\alpha(x)$ by solving (18). For the Tanh model, $\int_{S_0}^{S} \mathrm{d}y/(y\,\eta(y))$ can be computed numerically. Then $G_\alpha(x) = x\,\eta^2(F_\alpha(x))$, and J_α is the Asian option rate function for the "local volatility" model with effective local volatility determined by G_α. This is computed using the methods of [2].

The optimization over α is a one-dimensional minimization of $C(\alpha, \rho) \cdot J_\alpha(V_0, K)$, which is performed numerically.

6.2 Results

Table 1 compares our alpha-optimal bound with the existing bounds from [1] for OTM variance options across a range of correlations and strikes. The results demonstrate substantial improvement, particularly for negative ρ (the financially relevant leverage regime) and out-of-the-money calls ($K > F_V(0)$).

6.3 Structure of the optimal coupling parameter

A striking feature of the numerical results is the behavior of the optimal $\alpha^*(\rho, K)$. Table 1 shows that:

1. The optimal α^* is consistently shifted toward *negative* values relative to ρ, reflecting the leverage effect in the Tanh model (η is decreasing).

Table 1: Comparison of upper bounds for the rate function I_ρ of variance call options in the Tanh local volatility model (29). "P13" = Proposition 13 of [1], equal to $J(V_0, K/\eta_0^2)/(\sigma_0^2(1-\rho^2))$. "$\alpha$-opt" = our optimized alpha-parametric bound (Theorem 3.3). "α^*" = optimal coupling parameter. "Impr" = percentage by which α-opt improves (reduces) P13. All entries are upper bounds for I_ρ.

ρ	P13	α-opt	α^*	Impr
$K = 1.2$ (OTM call, $x = 0.182$)				
-0.7	0.0236	0.0060	-0.89	74.4%
-0.5	0.0160	0.0067	-0.80	58.0%
-0.3	0.0132	0.0076	-0.70	42.4%
$+0.3$	0.0132	0.0125	-0.26	5.2%
$+0.5$	0.0160	0.0160	-0.04	0.0%
$+0.7$	0.0236	0.0224	$+0.26$	5.1%
$K = 1.5$ (OTM call, $x = 0.405$)				
-0.7	0.1120	0.0280	-0.89	75.0%
-0.5	0.0762	0.0313	-0.81	59.0%
-0.3	0.0628	0.0354	-0.71	43.7%
$+0.3$	0.0628	0.0587	-0.29	6.4%
$+0.5$	0.0762	0.0760	-0.08	0.3%
$+0.7$	0.1120	0.1089	$+0.20$	2.7%
$K = 2.0$ (deep OTM call, $x = 0.693$)				
-0.7	0.3119	0.0765	-0.89	75.5%
-0.5	0.2121	0.0852	-0.80	59.8%
-0.3	0.1748	0.0963	-0.70	44.9%
$+0.3$	0.1748	0.1608	-0.32	8.0%
$+0.5$	0.2121	0.2102	-0.14	0.9%
$+0.7$	0.3119	0.3108	$+0.11$	0.4%

2. For OTM calls ($K > F_V(0)$) with negative ρ, the improvement over existing bounds reaches 75%, demonstrating that the alpha-parametric framework captures correlation structure that the fixed-α bounds miss entirely.

3. The improvement is asymmetric in ρ: negative correlation (leverage) yields much larger gains than positive correlation. This is because the square-root coupling in (18) interacts non-trivially with the decreasing local volatility, and the optimal trajectory exploits this interaction when $\rho < 0$.

4. For OTM puts ($K < F_V(0)$) with positive ρ, the alpha-optimal bound can be *worse* than Proposition 13 (negative improvement), because the square-root coupling is not the natural parameterization for this regime. In such cases, the logarithmic coupling of Proposition 17 of [1] may be preferred.

These observations suggest that in practice, one should compute both the alpha-optimal bound and the Proposition 17 bound of [1], and take the minimum. This combined bound is never worse than either individual bound.

7 Combined Bound and Comparison with Logarithmic Coupling

The numerical experiments reveal that the alpha-parametric (square-root coupling) bound excels for negative ρ with OTM calls, while the Proposition 17 bound of [1] (logarithmic coupling) may be better for other parameter regimes. We formalize this observation.

Definition 7.1. *The* combined bound *is*

$$I_\rho^{\text{comb}}(S_0, V_0, K) := \min\Big\{ \inf_{\alpha\in\mathbb{R}} C(\alpha,\rho)\cdot J_\alpha^{\text{sqrt}}(V_0,K),\ J_\rho^{\log}(V_0,K)\Big\}, \tag{30}$$

where J_α^{sqrt} uses the square-root coupling (18) *and $J_\rho^{\log}$ uses the logarithmic coupling of Proposition 17 in [1].*

Proposition 7.2 (Combined bound)**.** *For any $\rho \in (-1,1)$ and $K>0$:*

1. *$I_\rho \le I_\rho^{\text{comb}} \le \min\big\{ I_0/(1-|\rho|),\ J(V_0, K/\eta_0^2)/(1-\rho^2)\big\}$.*
2. *When η is constant, $I_\rho^{\text{comb}} = I_\rho = J(V_0, K/\eta_0^2)$.*

Proof. Part (1): both individual bounds in (30) are valid upper bounds for I_ρ (the square-root bound by Theorem 3.3, the logarithmic bound by Proposition 17 of [1]), and their minimum is at most as large as either. Part (2) follows from Corollary 3.7 and the analogous result for the logarithmic coupling (Remark 15 of [1]). □

Remark 7.3. *The numerical experiments in Table 1 suggest that the square-root coupling is tighter for $\rho<0$ with OTM calls ($K > F_V(0)$), while the logarithmic coupling of [1] may be tighter in other regimes. The optimal choice depends on model parameters, motivating the combined bound.*

8 Conclusion

Limitations. Our results are derived under the assumption $\sigma(v)\equiv\sigma_0$, which merits discussion. This assumption is essential for Proposition 2.1 (the Hamiltonian conservation requires time-autonomy of the Lagrangian) and simplifies the alpha-parametric bound (the velocity $b = h'/\sigma_0$ becomes a pure derivative). For the perturbation expansion (Theorem 4.2), the constant-σ assumption is used but not essential: the envelope theorem argument applies identically with general $\sigma(v)$, though the explicit form of I_1 would involve the non-constant $\sigma(e^{h_0(t)})$. In practice, the constant-σ restriction excludes models such as the 3/2 model ($\sigma(v)=\sigma v^{-1/2}$) and the SABR-type vol-of-vol. Extending the alpha-parametric framework to non-constant σ requires replacing the velocity variable $b=h'/\sigma_0$ with $b=h'/\sigma(e^h)$, which introduces h-dependent coefficients into the action and prevents the clean factorization in Theorem 3.3. We leave this extension for future work.

We have developed three complementary approaches to the open problem of computing the rate function I_ρ for short-maturity variance options in LSV models with non-zero correlation. The alpha-parametric bound provides a practical, computable approximation

that is strictly tighter than all existing bounds and exact in the constant local volatility case. The perturbation expansion gives explicit formulas for small $|\rho|$ with clear financial interpretation. The Hamiltonian structure reveals the underlying mechanical nature of the optimization problem.

Several extensions are natural. First, the case of general $\sigma(v)$ (non-constant vol-of-vol) can be approached by incorporating the σ' terms into the Hamiltonian analysis. Second, connections to rough volatility models, where the Volterra kernel replaces the SDE dynamics, could be explored via the sample-path LDP framework of [5]. Third, options on the newly launched CBOE Variance Futures provide an empirical testing ground for our asymptotic formulas.

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